

Data Analysis

NETFLIX MOVIE ANALYSIS

With PostgreSQL



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Hi. I'm Zaky!

Junior Data Analyst

HALOHA!

This project was specifically my third project in SQL. This time, I got curious about analysis of **Netflix Title Dataset**.

Before really jump into the project, let me share some summary of how this project was done.

The Netflix Movie Analysis was conducted by analyzing a dataset from **Kaggle**. Dataset was pre-processed through **Jupyter Notebook** (Python) then it was connected to **PostgreSQL** for cleaning, transformation, and analyzing.

Cleaning process was done mainly on removing duplicates, handling null values, correcting misplaced value, and reformatting column types. In **transformation process**, another additional tables were extracted and generated from raw data to provide more precise information for further analysis.

Lastly, several question was raised regarding cleaned dataset to gain meaningful information on **directors and genre productivity**..

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At the beginning, I challenged myself to import the dataset using Python and perform additional preprocessing before further handling it in PostgreSQL.

The netflix_raw table contains 12 columns: 'show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added', 'release_year', 'rating', 'duration', 'listed_in', and 'description'.

Package and dataset successfully imported

When nan appears, that might be

[23]: 807

Cleaning Process (PostgreSQL)

After being imported, several cleaning methods were conducted. The aim was to eliminate anomalous entries like duplicated rows/values, misplaced values, unstandardized format, and null values.

- Duplicated values/rows were excluded in netflix_cleaned table.
- Misplaced values in duration were taken back to their appropriate column.
- Date column was set to proper date type.
- Null value in country column were populated using the country value associated with the same director (previously available in another row).

etc.

Query Query History

28 -- Find any null value for each column
29 SELECT
30 COUNT(*) FILTER(WHERE show_id IS NULL) AS show_id,
31 COUNT(*) FILTER(WHERE type IS NULL) AS type,
32 COUNT(*) FILTER(WHERE title IS NULL) AS title,
33 COUNT(*) FILTER(WHERE director IS NULL) AS director,
34 COUNT(*) FILTER(WHERE "cast" IS NULL) AS cast,
35 COUNT(*) FILTER(WHERE country IS NULL) AS country,
36 COUNT(*) FILTER(WHERE date_added IS NULL) AS date_added,
37 COUNT(*) FILTER(WHERE release_year IS NULL) AS release_year,
38 COUNT(*) FILTER(WHERE rating IS NULL) AS rating,
39 COUNT(*) FILTER(WHERE duration IS NULL) AS duration,
40 COUNT(*) FILTER(WHERE listed_in IS NULL) AS listed_in,
41 COUNT(*) FILTER(WHERE description IS NULL) AS description
42 FROM netflix_raw
43 -- Found null values in director (2634), cast (825), country (832), date_added (10), rating (4),
44

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	des
	bigint	bigint	bigint	bigint	bigint	bigint	bigint	bigint	bigint	bigint	bigint	bigint
1	0	0	0	2634	825	831	10	0	7	0	0	

Total rows: 1 Query complete 00:00:00.283

Query Query History

51 -- Find duplicated row in similar show_id
52 SELECT
53 *
54 FROM netflix_raw
55 WHERE TRIM(LOWER(title)) IN (
56 SELECT
57 title
58 FROM (
59 SELECT
60 TRIM(LOWER(title)) AS title,
61 COUNT(TRIM(LOWER(title))) AS freq

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	show_id	type
	[PK] character varying (10)	character varying (10)
1	s5319	Movie
2	s5752	TV Show
3	s304	Movie
4	s6706	Movie

Total rows: 10 Query complete 00:00:00.509

165 -- Exchange data from both place.
166 UPDATE netflix_raw
167 SET
168 duration = rating,
169 rating = NULL
170 WHERE duration IS NULL
171 -- Null values in duration already resolved.
172
173
174 -- Reformatting date_added column from varchar to date
175 ALTER TABLE netflix_raw
176 ALTER COLUMN date_added TYPE DATE
177 USING TO_DATE(date_added, 'Month DD, YYYY')
178 -- Reformatting success from November 25, 2016
179

Transformation Process (PostgreSQL)

In this process, additional tables were extracted and generated from `netflix_raw` to provide more precise information for every need. These tables include **`netflix_country`**, **`netflix_director`**, **`netflix_genre`**, **`netflix_cast`**, and **`netflix_cleaned`**.

All of those were essential for supporting the cleaning process and performing further analysis.

```
Query  Query History
104  -- Assign different table dedicated for show_id with (single) director, listed_in,
105  -- Assign cast
106  SELECT
107      show_id,
108      TRIM(UNNEST(STRING_TO_ARRAY("cast", ','))) AS cast_name
109  INTO netflix_cast
110  FROM netflix_raw
111  -- Successfully attempt 64126 entries
112
113  -- Assign listed_in
114  SELECT
115      show_id,
116      TRIM(UNNEST(STRING_TO_ARRAY(listed_in, ','))) AS genre_name
117  INTO netflix_genre
118  FROM netflix_raw
119  -- Successfully attempt 19323 entries
120
121  -- Assign country
122  SELECT
```

Tables (6)

- > netflix_cast
- > netflix_clean
- > netflix_country
- > netflix_director
- > netflix_genre
- ▼ netflix_raw

```
Query  Query History
181  -- Retrieve main columns for analysis, excluding duplicated rows. Generate new table of it.
182  WITH helper AS (
183  SELECT
184      *,
185      ROW_NUMBER() OVER (PARTITION BY LOWER(title), type ORDER BY show_id) AS rn
186  FROM netflix_raw
187  )
188  SELECT
189      show_id,
190      type,
191      title,
192      date_added,
193      release_year,
194      rating,
195      duration,
196      description
197  INTO netflix_clean
198  FROM helper
199  WHERE rn = 1
```

```

2  SELECT
3      nd.director_name,
4      COUNT(CASE WHEN nc.type = 'Movie' THEN 1 END) AS movies,
5      COUNT(CASE WHEN nc.type = 'TV Show' THEN 1 END) AS tv_shows,
6      COUNT(*) AS total
7  FROM netflix_director AS nd
8  JOIN netflix_clean AS nc
9  ON nd.show_id = nc.show_id
10 GROUP BY 1
11 HAVING COUNT(CASE WHEN nc.type = 'Movie' THEN 1 END) > 0
12 AND COUNT(CASE WHEN nc.type = 'TV Show' THEN 1 END) > 0
13 ORDER BY 4 DESC

```

For each director, count the number of Movies and TV shows created by the director who has directed both!

- Successfully retrieved 83 rows.
- Marcus Raboy was a director who directing film the most (15 movies and 1 tv show), followed by Anurag Kashyap (8 + 1).
- Approximately 44% of them have directed more than two films.

	director_name text	movies bigint	tv_shows bigint	total bigint
1	Marcus Raboy	15	1	16
2	Anurag Kashyap	8	1	9
3	Quentin Tarantino	7	1	8
4	Ryan Polito	7	1	8
5	Priyadarshan	6	1	7
6	Michael Simon	5	1	6
7	Abhishek Chaubey	4	1	5
8	Robb Wells	4	1	5
9	Oliver Stone	4	1	5
10	Joe Berlinger	3	2	5
11	Ken Burns	2	3	5
Total rows: 83		Query complete 00:00:00.374		


```

33 GROUP BY nc.country_name
34 ORDER BY comedy_number DESC
35 LIMIT 10
36 -- Make it precise. Devide by detailed comedy genre!
37 SELECT
38     nc.country_name,
39     COUNT(CASE WHEN ng.genre_name LIKE 'Comedies' THEN ng.genre_name ELSE NULL END) AS "Comedies",
40     COUNT(CASE WHEN ng.genre_name LIKE 'Stand-Up Comedy & Talk Shows' THEN ng.genre_name ELSE NULL END) AS "Stand-Up Comedy & Talk Shows",
41     COUNT(CASE WHEN ng.genre_name LIKE 'Stand-Up Comedy' THEN ng.genre_name ELSE NULL END) AS "Stand-Up Comedy",
42     COUNT(CASE WHEN ng.genre_name LIKE 'TV Comedies' THEN ng.genre_name ELSE NULL END) AS "TV Comedies",
43     COUNT(CASE WHEN ng.genre_name LIKE '%Comed%' THEN ng.genre_name ELSE NULL END) AS total
44 FROM netflix_genre AS ng
45 JOIN netflix_country AS nc
46 ON ng.show_id = nc.show_id
47 GROUP BY nc.country_name
48 ORDER BY total DESC

```

Which country has the highest number of comedy films?

After assessing, there were several keywords associated with comedy in the genre column. This information was important to figure out the answer.

- Only 71 out of 123 countries have comedy films in the dataset. Even half of them have fewer than 10 comedy films.
- The US (1,200) placed first in terms of comedy genre production, followed by India (371) and The UK (160).

	country_name text	Comedies bigint	Stand-Up Comedy & Talk Shows bigint	Stand-Up Comedy bigint	TV Comedies bigint	total bigint
1	United States	685	33	224	258	1200
2	India	336	3	6	26	371
3	United Kingdom	92	1	23	44	160
4	Canada	95	0	2	30	127
5	France	51	1	5	24	81
6	Egypt	63	0	0	4	67
7	Turkey	58	0	0	3	61
8	Spain	49	0	2	5	56
9	Germany	42	1	5	5	53
10	Mexico	27	1	18	4	50
11	Nigeria	11	0	0	2	16
Total rows: 123 Query complete 00:00:00.182						

For each year, which director has maximum number of movies released?

After retrieving and sorting directors list by show_id and release_year, the list was grouped and ranked by the number of films per director and year, so only the director who placed first would appeared.

- Successfully retrieved the name of the most productive director per year (73 years from 1942-2021).
- For the last 10 years (as per record), Jan Suter (2018) made the highest number of films among any other director who only made 5-6 films per year.

```
63 -- Grouping rows per director and release_year and ranking it inside the CTE.
64 WITH year_ranking AS (
65     SELECT
66         nd.director_name,
67         nc.release_year,
68         COUNT(*) AS freq,
69         ROW_NUMBER() OVER (PARTITION BY nc.release_year ORDER BY COUNT(*) DESC, nd.director_name)
70     FROM netflix_director AS nd
71     JOIN netflix_clean AS nc
72     ON nd.show_id = nc.show_id
73     GROUP BY 1, 2
74     ORDER BY 2 DESC, 4
75 )
76 SELECT
77     release_year,
78     director_name,
79     freq
80 FROM year_ranking
81 WHERE rn = 1
82 ORDER BY 1 DESC
```

	release_year 	director_name 	freq 
	integer	text	bigint
1	2021	Rathindran R Prasad	5
2	2020	Alex Woo	4
3	2019	Fernando Ayllón	4
4	2018	Jan Suter	12
5	2017	Marcus Raboy	6
6	2016	Jan Suter	4
7	2015	S.S. Rajamouli	4
8	2014	Jay Chapman	3
9	2013	Rajiv Chilaka	6
10	2012	Rajiv Chilaka	4
11	2011	Rajiv Chilaka	5
Total rows: 73		Query complete 00:00:00.184	


```

101 WITH avg_durations AS (
102     SELECT
103         ng.genre_name,
104         nc.type,
105         ROUND(AVG(SPLIT_PART(duration, ' ', 1)::INT), 2) AS avg_duration
106     FROM netflix_genre AS ng
107     JOIN netflix_clean AS nc
108     ON ng.show_id = nc.show_id
109     GROUP BY 1, 2
110 )
111 SELECT
112     *
113 FROM avg_durations
114 WHERE type = 'Movie'
115 ORDER BY avg_duration DESC;

```

What is the average duration for each film across genres?

For this question, the durations for each genre was averaged and sorted in descending number. The averaged duration then formatted as short decimal.

- The longest duration came from The Classical Movies genre with 118 minutes.
- The shortest can be found in The Movie Movies genre (47 minutes) and Stand-Up Comedy Movies (67 minutes) genre.

	genre_name text	type character varying (10)	avg_duration numeric
1	Classic Movies	Movie	118.65
2	Action & Adventure	Movie	113.52
3	Dramas	Movie	113.06
4	Romantic Movies	Movie	110.57
5	International Movies	Movie	110.36
6	Thrillers	Movie	107.17
7	Music & Musicals	Movie	106.13
8	Sci-Fi & Fantasy	Movie	106.02
9	Faith & Spirituality	Movie	105.58
10	Cult Movies	Movie	104.66
11	Comedies	Movie	102.88
Total rows: 20		Query complete 00:00:00.095	

```

131 WITH horror_comedy AS (
132     SELECT
133         nd.director_name,
134         ng.genre_name
135     FROM netflix_director AS nd
136     JOIN netflix_genre AS ng
137     ON nd.show_id = ng.show_id
138     WHERE ng.genre_name IN ('Horror Movies', 'Comedies')
139 ),
140 count_genre AS (
141     SELECT
142         director_name,
143         COUNT(CASE WHEN genre_name = 'Horror Movies' THEN 1 END) AS horror,
144         COUNT(CASE WHEN genre_name = 'Comedies' THEN 1 END) AS comedy
145     FROM horror_comedy
146     GROUP BY director_name
147 )
148 SELECT
149     *,
150     (horror + comedy) AS total
151 FROM count_genre
152 WHERE horror > 0 AND comedy > 0
153 ORDER BY total DESC

```

	director_name text	horror bigint	comedy bigint	total bigint
1	Kevin Smith	3	5	8
2	Poj Arnon	5	3	8
3	Don Michael Paul	3	3	6
4	McG	2	4	6
5	Michael Tiddes	2	4	6
6	Steve Brill	1	5	6
7	Zoya Akhtar	1	3	4
8	Scott Stewart	3	1	4
9	Ahmed Zein	2	2	4
10	Dibakar Banerjee	1	3	4
11	Parvaz Durrani	2	1	4
Total rows: 55		Query complete 00:00:00.109		

Find directors who already work on both horror movies and comedy movies!

Thanks to CTE queries, this problem can be done easily!

- Found 55 rows of director.
- Kevin Smith and Poj Arnon placed the top leaderboard (8 films).

What do you think? Check out more on link below!

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